# Ordinary Least Squares

**Lecture 2 (R code)**

## Table of contents


Reading . . . 2

**Data** **3**
LCF (2023) . . . 3

**Estimation: An intuitive approach** **3**
Guess 1 . . . 5
Guess 2 . . . 6
Structure . . . 7
Fit . . . 8
Aggregation . . . 8
Minization . . . 9

**Ordinary Least Squares** **9**
Problem . . . 9
First order conditions . . . 9
Simple case . . . 10
Sample Analogue . . . 11
Decomposition . . . 12
Fitted values . . . 12
Interpretation . . . 12

**Example: LCF (2023)** **13**
Estimation . . . 13
Decomposition . . . 13
Fitted values . . . 14
Residual . . . 15
Binary regressor: . . . 16
Summary stats . . . 16

OLS - dummy variable . . . 17
Sample analogue . . . 18
Multivariate models . . . 18
OLS - multivariate . . . 18
Interpretation . . . 19
Partialling out . . . 19
Demonstration . . . 20

**Goodness-of-Fit** **21**
Measure of fit . . . 22
Chasing $\mathbf{R}^2$ . . . 22
Adjusted $\mathbf{R}^2$ . . . 23
Example: LCF (2023) . . . 23
Adding "irrelevant" variables . . . 23

**Does OLS work?** **24**
Accuracy . . . 26
Random variable . . . 26
Example: LCF (2023) - Version 2 . . . 26
Example: LCF (2023) - Version 3 . . . 27
Example: LCF (2023) - Version 4 . . . 27

**References** **28**
Bibliography . . . 28

## Reading

This lecture covers material from Wooldridge (2025):

- Chapter 2.2 - "Deriving the Ordinary Least Squares Estimates"
- Chapter 2.3 - "Properties of OLS on Any Sample of Data"
- Chapter 2.4 - "Units of Measurement and Functional Form" (covered in Lecture 1)
- Chapter 3.2 - "Mechanics and Interpretation of Ordinary Least Squares"
- Chapter 6.3 - "More on Goodness-of-Fit and Selection of Regressors"

The lecture does not cover Chapter 2.6 "Regression through the Origin and Regression on a Constant". I would suggest reading the section, but we will not have time to cover it.

# Data

For each observation we observe:

$$[y_i, x_{i1}, x_{i2}, \dots, x_{ik}] \quad i = 1, \dots, n$$

Stacked together, we have a matrix of data:

$$\begin{bmatrix} y_1 & x_{11} & x_{12} & \dots & x_{1k} \\ y_2 & x_{21} & x_{22} & \dots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ y_n & x_{n1} & x_{n2} & \dots & x_{nk} \end{bmatrix}$$

Just like in Stata/R

## LCF (2023)

Suppose you wanted to estimate the the marginal propensity to consume in the UK using data from the Living Costs and Food Survey:

- a *household level* dataset with information on consumption, income, and other characteristics.

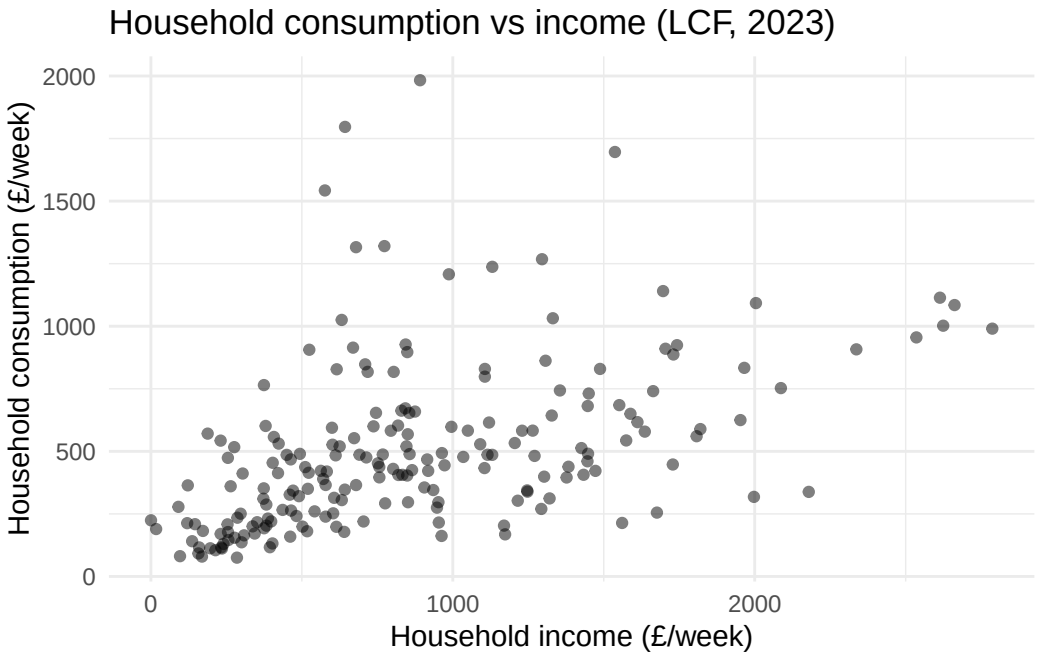
$$[cons_i, inc_i, children_i, size_i, tv_license_i, month_i] \quad i = 1, \dots, n$$

# Estimation: An intuitive approach

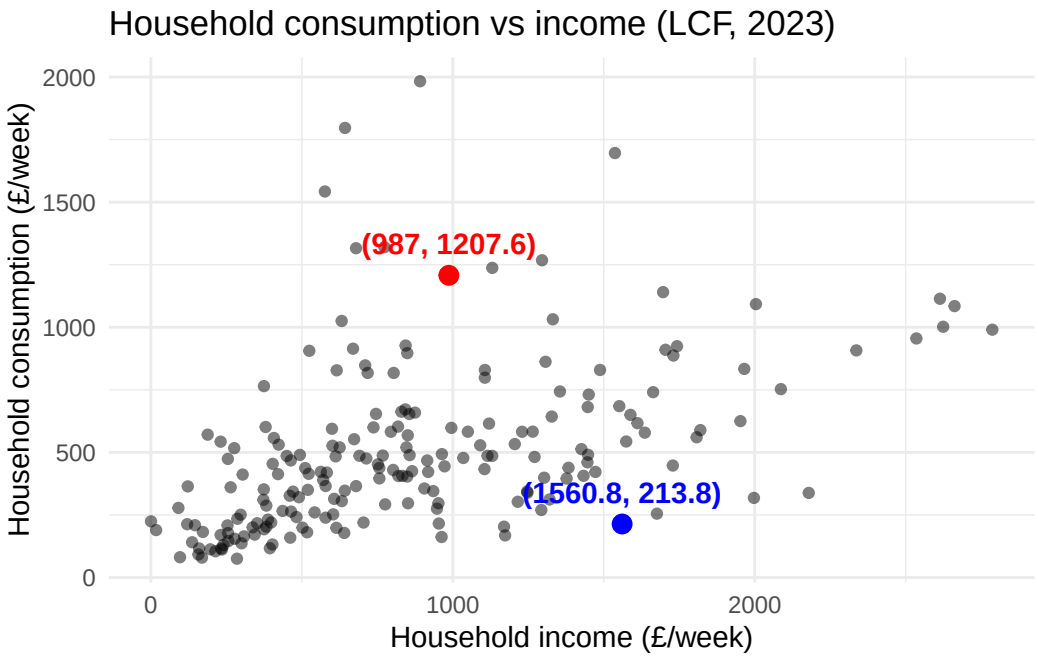
Does the relationship between consumption and income look linear?

```r
# Load the Stata file
lcf_data <- read_dta(file.path(data_dir, "lcf_2023.dta"))

# Plot
ggplot(lcf_data, aes(x = hhinc, y = hhcon)) +
  geom_point(alpha = 0.5) +
  labs(x = "Household income (£/week)",
       y = "Household consumption (£/week)",
       title = "Household consumption vs income (LCF, 2023)") +
  theme_minimal()
```

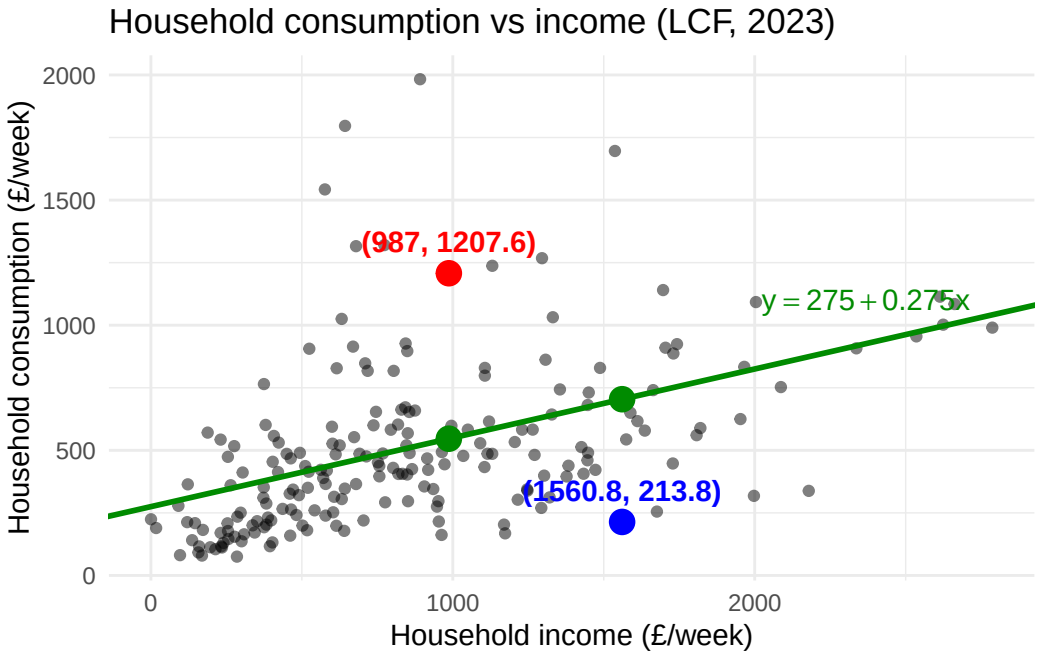



Consider the following two observations:

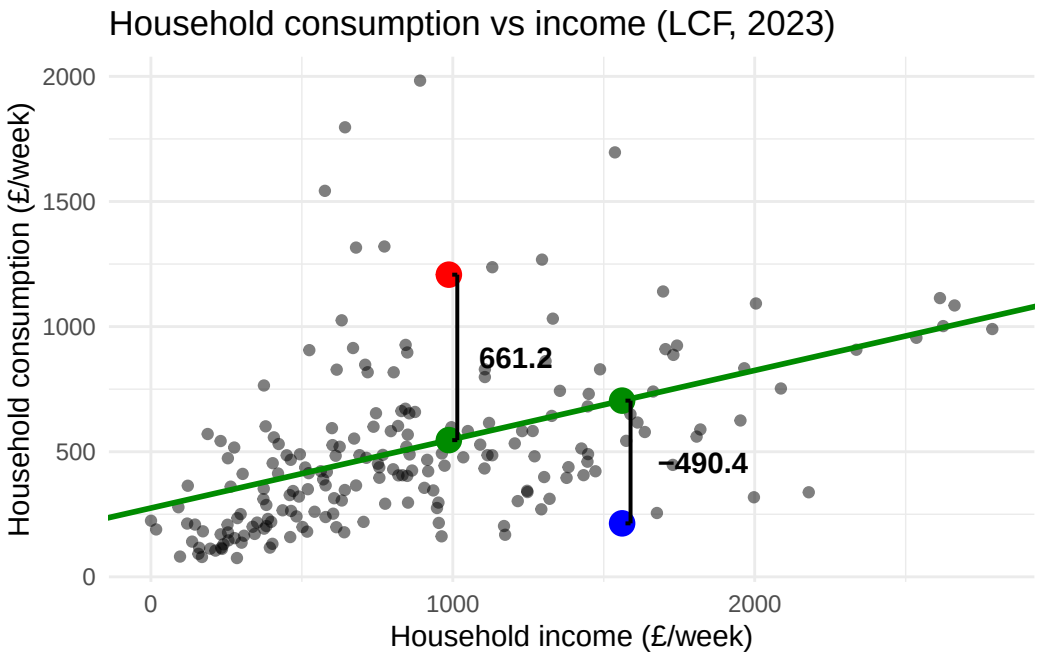



### Guess 1

Suppose we plot a line through the points:

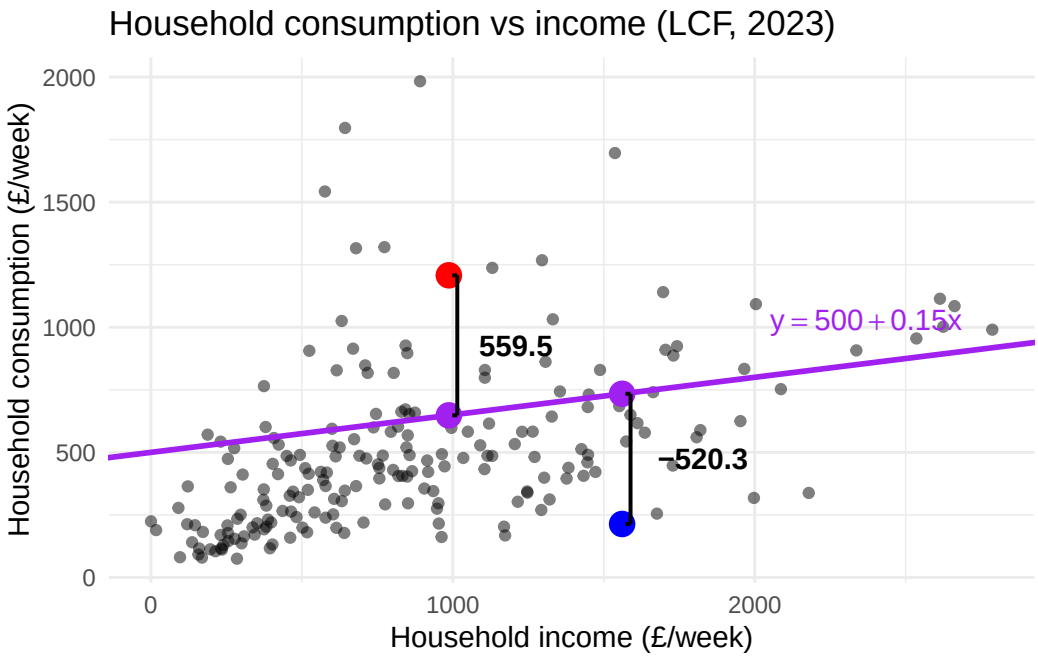


The vertical distance between points in the data and the line tells us how the difference between the "prediction" of the line (for a given $inc_i$) and the actual data.




## Guess 2

What if we chose a different plot?




**Which guess was better? Why?**

## Structure

Let's formalize the above process:

- **Goal:** we want to pick the line that best fits the data.
- **Problem:** what do we mean by "fit"?
  - For a given value of $inc_i$ in the data, the "fitted" value of $cons_i$ is

$$\widehat{cons}_i = b_0 + b_1 inc_i$$

    where $[b_0, b_1]$ are the chosen parameters.
  - For example,
    * $\widehat{cons}_i = 275 + 0.275 inc_i$
    * $\widehat{cons}_i = 500 + 0.15 inc_i$

## Fit

We need a measure of distance:

**Candidate 1**: difference

$$cons_i - \widehat{cons}_i$$

- **PROBLEM**: could be negative (i.e. not a measure of distance)

**Candidate 2**: absolute value

$$|cons_i - \widehat{cons}_i|$$

- **PRO**: intuitive
- **CON**: difficult to work with (i.e. not globally differentiable)

**Candidate 3**: squared deviation

$$(cons_i - \widehat{cons}_i)^2$$

- always positive
- amplifies bigger deviations, which is more *efficient* (more on that soon!)

## Aggregation

What we care about is the difference between the prediction and data across all observations:

$$\sum_{i=1}^{n}(cons_i - \widehat{cons}_i)^2$$

And since $\widehat{cons}_i = b_0 + b_1 inc_i$

$$\sum_{i=1}^{n}(cons_i - b_0 - b_1 inc_i)^2$$

### Minization

The value of these squared deviations (from the prediction) depend on $[b_0, b_1]$. It seems intuitive then to pick the values that minimize this value:

$$\min_{b_0,b_1} \quad \sum_{i=1}^{n}(cons_i - b_0 - b_1 inc_i)^2$$

## Ordinary Least Squares

### Problem

Minimization problem:

$$\hat{\beta} = \arg\min_{b} \quad \sum_{i=1}^{n}(y_i - b_0 - b_1 x_{i1} - \cdots - b_k x_{ik})^2$$

where $\hat{\beta} = [\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k]$ and $b = [b_0, b_1, \ldots, b_k]$.

- hence the name "Least Squares"

### First order conditions

Differentiate w.r.t. $b_j$'s and then set to 0. For $\hat{\beta}_0$ (the constant):

$$\begin{aligned} & 0 = 2\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) \\ \Rightarrow \quad & 0 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) \\ \Rightarrow \quad & \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}_1 - \cdots - \hat{\beta}_k \bar{x}_k \end{aligned}$$

> **Note**
>
> The inclusion of a constant in the model ensures that the residual is always mean zero:
>
> $$0 = \sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) = \sum_{i=1}^{n}\hat{u}_i$$

For $\hat{\beta}_j \quad j = 1, \ldots, k$

$$\begin{aligned} & 0 = 2\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik})x_{ij} \\ \Rightarrow \quad & 0 = \sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik})x_{ij} \end{aligned}$$

There are $k$ of these first-order conditions (plus the 1 for the constant)

$\Rightarrow k + 1$ equations and $k + 1$ unknowns.

## Simple case

In the case of a single regressor, the solution is more easily solved.

- For $\hat{\beta}_0$ (the constant):

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

- For $\hat{\beta}_1$:

$$\begin{aligned} & 0 = 2\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)x_i \\ \Rightarrow \quad & 0 = \sum_{i=1}^{n}(y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) - \hat{\beta}_1 x_i)x_i \qquad \text{substitute in} \quad \hat{\beta}_0 \\ \Rightarrow \quad & 0 = \sum_{i=1}^{n}((y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x}))x_i \end{aligned}$$

The next step exploits the fact that:

$$\sum_{i=1}^{n} \hat{u}_i = 0 \Rightarrow \bar{x} \sum_{i=1}^{n} \hat{u}_i = 0 \Rightarrow \sum_{i=1}^{n} \hat{u}_i \bar{x} = 0$$

Therefore we can minus 0 from the right-hand side without changing anything.

$$\begin{aligned}
0 &= \sum_{i=1}^{n} \big( \underbrace{(y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x})}_{\hat{u}_i} \big) x_i - \sum_{i=1}^{n} \hat{u}_i \bar{x} \\
\Rightarrow \quad 0 &= \sum_{i=1}^{n} \big( (y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x}) \big) (x_i - \bar{x}) \\
\Rightarrow \quad \hat{\beta}_1 &= \frac{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} = \frac{\widehat{Cov}(y_i, x_i)}{\widehat{Var}(x_i)}
\end{aligned}$$

Linear algebra

The above expression - along with its multivariate counterpart - has a very nice linear algebra counterpart. The vector of OLS estimators (including the intercept) is given by,

$$\hat{\beta} = (X'X)^{-1} X'Y$$

Where $X$ is a $n \times (k+1)$ matrix of regressors (including vector of 1's for the intercept) and $Y$ a $n \times 1$ vector. It's a very simple expression and demonstrates the value of knowing a little linear algebra. This how you define the OLS estimator in graduate-level modules like EC5207 *Econometric Methods and Applications*.

## Sample Analogue

Recall from Lecture 1, under the MLR assumptions (notably, $E[x_i u_i] = 0$)

$$\begin{aligned}
\beta_1 &= \frac{Cov(y_i, x_i)}{Var(x_i)} \\
\beta_0 &= E[y_i] - \beta_1 E[x_i]
\end{aligned}$$

The OLS estimators for a simple model are the **sample analogue**

$$\hat{\beta}_1 = \frac{\widehat{Cov}(y_i, x_i)}{\widehat{Var}(x_i)}$$
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

OLS replaces expectations with sample averages.

## Decomposition

Given the OLS estimates, we can decompose the outcome into two parts: fitted values ($\hat{y}_i$) and residual ($\hat{u}_i$)

$$y_i = \underbrace{\hat{\beta}_0 + \hat{\beta}_1 x_i}_{\hat{y}_i} + \hat{u}_i$$

## Fitted values

You can show,

- $\sum_{i=1}^n \hat{u}_i = 0$: residuals are mean zero
- $\sum_{i=1}^n x_i \hat{u}_i = 0$: residuals and regressors are uncorrelated
- $\sum_{i=1}^n \hat{y}_i \hat{u}_i = 0$: residuals and fitted values are uncorrelated
- $\sum_{i=1}^n \hat{y}_i = \sum_{i=1}^n y_i$: fitted values have the same mean as outcome
- $\sum_{i=1}^n y_i \hat{u}_i = \sum_{i=1}^n (\hat{y}_i + \hat{u}_i)\hat{u}_i = \sum_{i=1}^n \hat{u}_i^2$

Some of these are exercises in Tutorial 1

## Interpretation

The $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ also gives us the interpration of the estimator:

$$\hat{\beta}_1 = \frac{d\hat{y}_i}{dx_i} \quad \text{OR} \quad \Delta\hat{y}_i = \hat{\beta}_1 \Delta x_i$$

and

$$\hat{\beta}_0 = \hat{y}_i \quad \text{for } x_i = 0$$

## Example: LCF (2023)

### Estimation

```r
# Estimate bivariate linear regression
model1 <- lm(hhcon ~ hhinc, data = lcf_data)

# View results
summary(model1)
```

```
Call:
lm(formula = hhcon ~ hhinc, data = lcf_data)

Residuals:
    Min      1Q  Median      3Q     Max
-519.70 -177.46  -60.08   95.78 1473.71

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 268.06105   37.42847   7.162 1.53e-11 ***
hhinc         0.27059    0.03582   7.555 1.52e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 291.3 on 198 degrees of freedom
Multiple R-squared:  0.2238,    Adjusted R-squared:  0.2198
F-statistic: 57.08 on 1 and 198 DF,  p-value: 1.515e-12
```

### Decomposition

```r
# Get fitted values and residuals
reg_vals <- data.frame(
  hhcon    = lcf_data$hhcon,
  fitted   = fitted(model1),
  resid    = residuals(model1)
)
```

```r
# Compute averages
colMeans(reg_vals)
```

```
       hhcon       fitted        resid
5.041663e+02 5.041663e+02 2.611245e-14
```

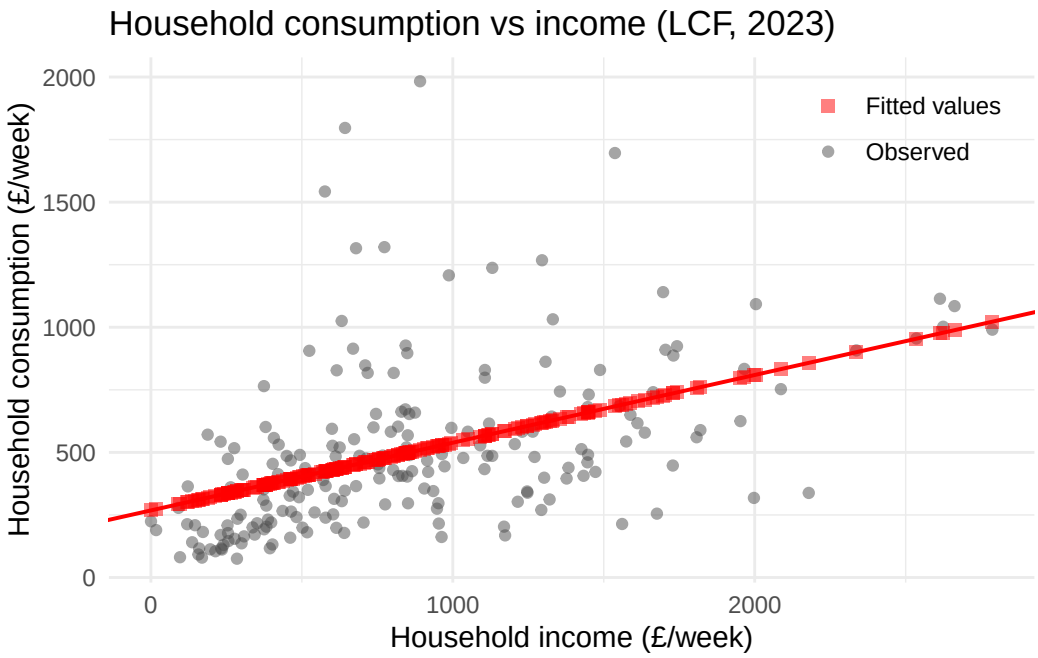
```r
# Compute variance-covariance matrix
var(reg_vals)
```

```
           hhcon       fitted        resid
hhcon  108747.43 2.433346e+04 8.441396e+04
fitted  24333.46 2.433346e+04 2.331285e-12
resid   84413.96 2.331285e-12 8.441396e+04
```

### Fitted values

```r
# Data frame with the x-variable plus fitted values and residuals
plot_vals <- data.frame(
  hhinc  = model.frame(model1)$hhinc,  # exactly the rows lm() used
  hhcon  = model.frame(model1)$hhcon,
  fitted = fitted(model1),
  resid  = residuals(model1)
)

# Plot
ggplot(plot_vals, aes(x = hhinc, y = hhcon)) +
  geom_point(aes(colour = "Observed"), alpha = 0.5) +
  geom_point(aes(y = fitted, colour = "Fitted values"), shape = 15, size = 2, alpha = 0.5) +
  geom_abline(intercept = coef(model1)[1], slope = coef(model1)[2],
              colour = "red", linewidth = 0.7) +
  scale_colour_manual(values = c("Observed" = "grey30",
                                 "Fitted values" = "red")) +
  labs(x = "Household income (£/week)",
       y = "Household consumption (£/week)",
       title = "Household consumption vs income (LCF, 2023)",
       colour = NULL) +
  theme_minimal() +
  theme(legend.position = c(0.98, 0.98),
        legend.justification = c(1, 1))
```

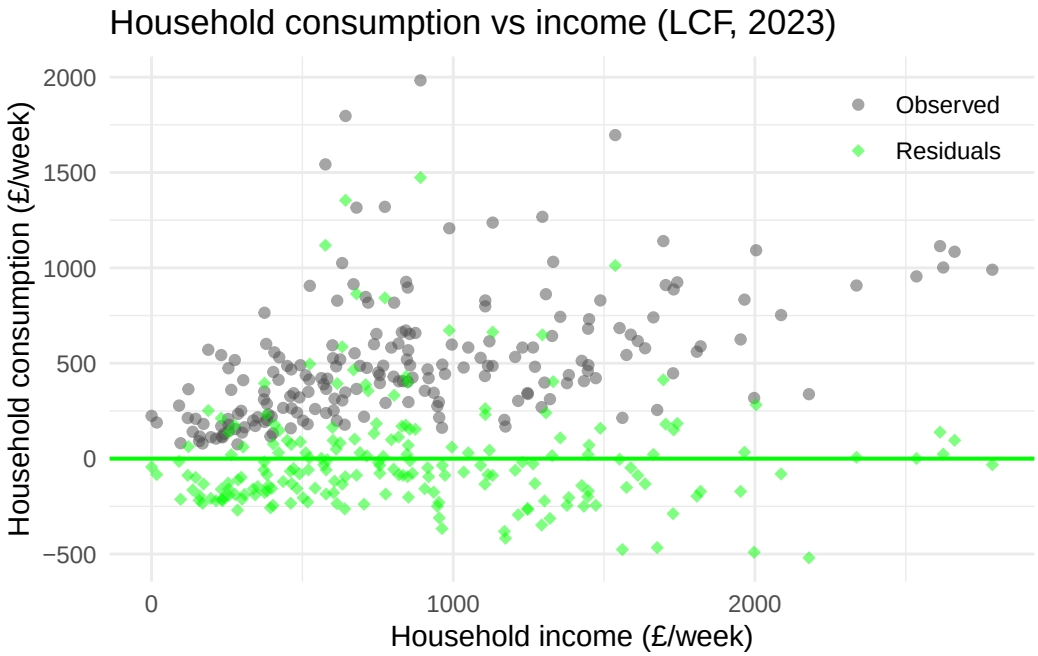



**Residual**

```
# Data frame with the x-variable plus fitted values and residuals
plot_vals <- data.frame(
  hhinc  = model.frame(model1)$hhinc,  # exactly the rows lm() used
  hhcon  = model.frame(model1)$hhcon,
  fitted = fitted(model1),
  resid  = residuals(model1)
)

# Plot
ggplot(plot_vals, aes(x = hhinc, y = hhcon)) +
  geom_point(aes(colour = "Observed"), alpha = 0.5) +
  geom_point(aes(y = resid, colour = "Residuals"), shape = 18, size = 2, alpha = 0.5) +
  geom_hline(yintercept = 0, colour = "green", linewidth = 0.7) +
  scale_colour_manual(values = c("Observed" = "grey30",
                                 "Residuals" = "green")) +
  labs(x = "Household income (£/week)",
       y = "Household consumption (£/week)",
       title = "Household consumption vs income (LCF, 2023)",
       colour = NULL) +
  theme_minimal() +
```

```
  theme(legend.position = c(0.98, 0.98),
        legend.justification = c(1, 1))
```




### Binary regressor:

What happens if the regressor is a 'dummy' variable?

$$[hhcons_i, children_i]$$

where $children_i$

- $= 1$: household has children,
- $= 0$: household does not have children

### Summary stats

```
# Summary table
tapply(lcf_data$hhcon, lcf_data$children, mean, na.rm = TRUE)
```

```
       0        1
481.0979 575.2546
```

## OLS - dummy variable

```r
# Estimate bivariate linear regression
model_dummy <- lm(hhcon ~ children, data=lcf_data)

summary(model_dummy)
```

```
Call:
lm(formula = hhcon ~ children, data = lcf_data)

Residuals:
    Min      1Q  Median      3Q     Max
-405.94 -248.78  -60.47  110.39 1501.92

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   481.10      26.70  18.019   <2e-16 ***
children       94.16      53.94   1.746   0.0824 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 328.1 on 198 degrees of freedom
Multiple R-squared:  0.01516,   Adjusted R-squared:  0.01018
F-statistic: 3.047 on 1 and 198 DF,  p-value: 0.08244
```

You can show,

$$\hat{\beta}_1 = \bar{y}_1 - \bar{y}_0$$
$$\hat{\beta}_0 = \bar{y}_0$$

where,

- $\bar{y}_0$ is the mean of the outcome for observations with $x = 0$
- $\bar{y}_1$ is the mean of the outcome for observations with $x = 1$

**Yields a very intuitive interpretation of the OLS estimator**

### Sample analogue

This is the sample analogue of

$$\begin{aligned}\beta_1 =& E[y_i|x_i = 1] - E[y_i|x_i = 0] \\ \beta_0 =& E[y_i|x_i = 0]\end{aligned}$$

Which was the interpretation of the population parameter in a model with a binary ('dummy') variable.

### Multivariate models

What happens if you have two regressors

$$[hhcons_i, hhinc_i, children, hhsize_i]$$

### OLS - multivariate

```
# Estimate bivariate linear regression
model_multi <- lm(hhcon ~ hhinc + children + hhsize, data=lcf_data)

summary(model_multi)
```

```
Call:
lm(formula = hhcon ~ hhinc + children + hhsize, data = lcf_data)

Residuals:
    Min      1Q  Median      3Q     Max
-456.92 -168.81  -58.49   96.59 1437.52

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  121.29566   47.91901   2.531  0.01215 *
hhinc          0.19372    0.03828   5.061 9.58e-07 ***
children    -238.35377   74.84546  -3.185  0.00169 **
hhsize       125.74181   27.57406   4.560 8.99e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 278.1 on 196 degrees of freedom
Multiple R-squared:  0.2995,    Adjusted R-squared:  0.2888
F-statistic: 27.93 on 3 and 196 DF,  p-value: 4.407e-15
```

### Interpretation

Interpretation follows from the fact that,

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \cdots + \hat{\beta}_k x_{ik}$$

$$\hat{\beta}_j = \frac{\partial \hat{y}_i}{\partial x_{ij}}$$

or

$$\Delta \hat{y}_i = \hat{\beta}_j \Delta x_{ij} \quad \text{holding other } x\text{'s fixed}$$

### Partialling out

The following result is called the **Frisch-Waugh Theorem** (Wooldridge 2025, 76)

The OLS estimate for $\hat{\beta}_1$ from the multivariate 'projection'

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \cdots + \hat{\beta}_k x_{ik}$$

is the same as the OLS estimate for $\hat{\gamma}_1$ simple 'projection'

$$\hat{\tilde{y}}_i = \hat{\gamma}_0 + \hat{\gamma}_1 \tilde{x}_{i1}$$

> **Projection**
>
> Why the phrase 'projection'? One interpretation of OLS is geometric. The estimator 'projects' the outcome ($Y$) in the (vector) space spanned by the regressors (columns of $X$).
>
> $$\hat{Y} = X\hat{\beta} = \underbrace{X(X'X)^{-1}X'}_{P_X} Y$$
>
> The predicted value of $Y$ is just a 'projection' of $Y$: $P_X Y$. $P_X$ is an idempotent projection matrix. The residual then is,

$$\hat{U} = Y - \hat{Y} = Y - P_X Y = (\underbrace{I - P_X}_{M_X})Y$$

$M_X$ is another idempotent projection matrix. And crucially, $M_X P_X = 0$: the two projections are orthogonal. So, OLS projects Y into two vector spaces: one spanned ('explained by') $X$ and another orthogonal to that space. The predicted values and residuals are mechanically orthogonal, which means that they are perfectly uncorrelated *in the data*.

where,

- $\tilde{x}_{i1}$ is the residual from the regression of $x_1$ against all other regressors

$$\tilde{x}_{i1} = x_{i1} - (\hat{\rho}_0 + \hat{\rho}_1 x_{i2} + \cdots + \hat{\rho}_{k-1} x_{ik})$$

and,

- $\tilde{y}_i$ is the residual from the regression of $y$ against all regressors *excluding* $x_1$

$$\tilde{y}_i = y_i - (\hat{\psi}_0 + \hat{\psi}_1 x_{i2} + \cdots + \hat{\psi}_{k-1} x_{ik})$$

### Demonstration

```
# Estimate bivariate linear regression
model_f1 <- lm(hhinc ~ children + hhsize, data=lcf_data)
model_f2 <- lm(hhcon ~ children + hhsize, data=lcf_data)

frisch <- data.frame(
  hhinc_tilde  = residuals(model_f1),
  hhcon_tilde  = residuals(model_f2)
)

model_frisch <- lm(hhcon_tilde ~ hhinc_tilde, data = frisch)
summary(model_frisch)
```

```
Call:
lm(formula = hhcon_tilde ~ hhinc_tilde, data = frisch)

Residuals:
    Min      1Q  Median      3Q     Max
-456.92 -168.81  -58.49   96.59 1437.52

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.673e-14  1.957e+01   0.000        1
hhinc_tilde  1.937e-01  3.809e-02   5.086 8.44e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 276.7 on 198 degrees of freedom
Multiple R-squared:  0.1156,    Adjusted R-squared:  0.1111
F-statistic: 25.87 on 1 and 198 DF,  p-value: 8.437e-07
```

## Goodness-of-Fit

**How well did we fit the data?**

We have decomposed each observation into two parts:

$$y_i = \underbrace{\widehat{y}_i}_{\text{fitted values}} + \underbrace{\widehat{u}_i}_{\text{residual}}$$

We can apply this decomposition to the *total sum of squares* (SST)

$$\text{SST} \equiv \sum_{i=1}^{n} (y_i - \bar{y})^2$$

- The SST is a measure of the total variation in $y$.

We can show that (see Wooldridge 2025, 33–34),

$$\text{SST} = \text{SSE} + \text{SSR}$$

where,

- *explained sum of squares* (SSE):

$$\text{SSE} \equiv \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

- *residual sum of squares* (SSR):

$$\text{SSR} \equiv \sum_{i=1}^{n} \hat{u}_i^2$$

## Measure of fit

One key measure of fit then is:

$$\mathbf{R}^2 = \frac{\text{SSE}}{\text{SST}} = 1 - \frac{\text{SSR}}{\text{SST}}$$

This measure captures the share of the variance in the outcome variable explained by the regressors.

- If $\mathbf{R}^2 = 1$ then regressors perfectly predict the outcome.

## Chasing $\mathbf{R}^2$

Many students make the mistake of thinking that higher $\mathbf{R}^2$ is always better. Here are reasons this need not be the case:

1. In social sciences, the DGP is often very complex and we don't observe many explanatory variables.
    - given the same set of $\mathbf{x}$'s, different outcomes may have very different $\mathbf{R}^2$
2. If we care about the relationship between $y$ and a specific $x_j$, we do not necessarily care about explaining all of $y$.

3. We can always add variables and (mechanically) increase $\mathbf{R}^2$; however, this comes at a potential cost:

    a. we *MAY* inadvertantly increase the variance of our estimates (e.g., *irrelevant* variables)

    b. we *MAY* end up adding "bad controls"

## Adjusted $\mathbf{R}^2$

An alternative measure fit is the *adjusted* $\mathbf{R}^2$:

$$\overline{\mathbf{R}}^2 = 1 - \frac{\text{SSR}/(n-k-1)}{\text{SST}/(n-1)} = 1 - (1-\mathbf{R}^2) \times \frac{(n-1)}{(n-k-1)}$$

where,

- $n$ is sample size
- $k$ is the number of regressors $\Rightarrow k+1$ is the number of parameters estimated (including intercept).

As you increase $k$, you decrease $\overline{\mathbf{R}}^2$

- Creates a disincentive to add irrelevant regressors that do not also reduce $SSR$.

## Example: LCF (2023)

```
[1] 0.2994804
```

```
[1] 0.2887582
```

## Adding "irrelevant" variables

```
model_irrel <- lm(hhcon ~ hhinc + hhsize + children + month + tvlicense, data = lcf_data)
summary(model_irrel)
```

```
Call:
lm(formula = hhcon ~ hhinc + hhsize + children + month + tvlicense,
    data = lcf_data)

Residuals:
    Min      1Q  Median      3Q     Max
-468.33 -163.40  -53.35   84.28 1434.32

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   82.86154   77.50891   1.069  0.28637
hhinc          0.19118    0.03853   4.961 1.53e-06 ***
hhsize       124.16104   27.71213   4.480 1.27e-05 ***
children    -233.00879   75.21666  -3.098  0.00224 **
month         -1.27623    6.23481  -0.205  0.83803
tvlicense     59.82271   56.57194   1.057  0.29162
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 278.7 on 194 degrees of freedom
Multiple R-squared:  0.3035,    Adjusted R-squared:  0.2856
F-statistic: 16.91 on 5 and 194 DF,  p-value: 7.335e-14
```

## Does OLS work?

Are these the "right" numbers?

```
Call:
lm(formula = hhcon ~ hhinc, data = lcf_data)

Residuals:
    Min      1Q  Median      3Q     Max
-519.70 -177.46  -60.08   95.78 1473.71

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 268.06105   37.42847   7.162 1.53e-11 ***
hhinc         0.27059    0.03582   7.555 1.52e-12 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 291.3 on 198 degrees of freedom
Multiple R-squared:  0.2238,    Adjusted R-squared:  0.2198
F-statistic: 57.08 on 1 and 198 DF,  p-value: 1.515e-12
```

Let's evaluate the question:

1. it pre-supposes that there is a *right* answer (i.e. a correct estimate)
    - i.e., a *true* population parameter ($\beta$)
2. it implies a notion of "correctness"
    - do we mean $\beta = 0.27$?
    - do we mean $\beta \approx 0.27$?
    - do we mean $\beta$ is 'close to' 0.27?
3. it assumes that the OLS estimator can be *right*.

First off, let's clarify something:

$$\hat{\beta}$$

- the estimator

**...is a random variable**

$$0.27$$

- the estimate

**...is just a number**

## Accuracy

Is $\hat{\beta} = \beta$?

- No!
- In fact, the statement does not make sense since $\hat{\beta}$ is a random variable while $\beta$ is a non-random constant.
- If we are accurate, we can ask: $Pr(\hat{\beta} = \beta) = ?$
- The answer is: $Pr(\hat{\beta} = \beta|X) = 0$ for any $u$ with a continuous distribution (conditional on $X$: the sample of values of independent regressors).

## Random variable

In what sense is $\hat{\beta}$ a random variable?

- The realized value of $\hat{\beta}$ depends on the realized sample[1]
- In another 'state of the world', the sample may have realized differently

## Example: LCF (2023) - Version 2

I randomly sampled 200 observations from the LCF (2023) data, but could have equally have sampled a different 200 observations

```
Call:
lm(formula = hhcon ~ hhinc, data = lcf_data2)

Residuals:
    Min      1Q  Median      3Q     Max
-420.74 -150.66  -39.10   87.07 1763.32

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 232.65286   30.55968   7.613 1.07e-12 ***
hhinc         0.27121    0.02859   9.487  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

[1] Recall, if $u$ is independent of $\mathbf{x}$, then we can treat $\mathbf{x}$ as fixed. Therefore, the randomness (across samples) is really from different realizations of $u$. However, any discussion of $u$ and $\mathbf{x}$ pre-supposes a model which we have not yet defined.

```
Residual standard error: 242.3 on 198 degrees of freedom
Multiple R-squared:  0.3125,    Adjusted R-squared:  0.3091
F-statistic: 90.01 on 1 and 198 DF,  p-value: < 2.2e-16
```

### Example: LCF (2023) - Version 3

... and again,

```
Call:
lm(formula = hhcon ~ hhinc, data = lcf_data3)

Residuals:
    Min      1Q  Median      3Q     Max
-515.61 -149.84  -46.11  115.40 1483.04

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 189.52921   36.88368   5.139 6.61e-07 ***
hhinc         0.34613    0.03426  10.104  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 270.4 on 198 degrees of freedom
Multiple R-squared:  0.3402,    Adjusted R-squared:  0.3369
F-statistic: 102.1 on 1 and 198 DF,  p-value: < 2.2e-16
```

### Example: LCF (2023) - Version 4

... and again.

```
Call:
lm(formula = hhcon ~ hhinc, data = lcf_data4)

Residuals:
    Min      1Q  Median      3Q     Max
-450.45 -163.57  -75.46   70.08 3112.93

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 267.77370   48.63081   5.506 1.13e-07 ***
```

```
hhinc         0.24171    0.04585   5.271 3.53e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 356.8 on 198 degrees of freedom
Multiple R-squared:  0.1231,    Adjusted R-squared:  0.1186
F-statistic: 27.79 on 1 and 198 DF,  p-value: 3.526e-07
```

Can we **expect** – *a priori* – for $\hat{\beta}$ to be "on target"?

- This is a question of *biasedness*: is $E[\hat{\beta}] = \beta$?
- The answer: it depends!
- Specifically, it depends on the underlying **model**!

> **Note**
>
> It is an all to common mistake to conflate the estimator - in particular, the OLS estimator - with the linear regression model. People will say, "We estimate an OLS regression model". This doesn't really make sense, since you OLS is the estimator, not the name of the model. You can say, "We estimate a linear regression model using OLS".
> This may seem pedantic, but it is an important distinction since there are a range of estimators that can be used to estimate a linear regression model; all with different properties. In addition, the OLS estimator can be applied to any data, regardless of the true model. So it is a good idea to make sure your language is accurate.

## References

### Bibliography

Wooldridge, Jeffrey M. 2025. "Introductory Econometrics: A Modern Approach."